

	Course Syllabus PHYS 325 Mathematical Methods for Scientists and Engineers FALL 2026-27
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INSTRUCTOR	
Name	Taylan Akdoğan
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Phone Number	0216-564 99 59
Office Number	AB1.309
Office Hours	Friday 13:00-14:00 or by appointment
Research/Teaching Assistants	Gülşah Bıçkı – gulsah.bicki@ozu.edu.tr

COURSE ORGANIZATION	
Interaction Methods	LMS and email
Technology Used by the Instructor	Generative AI for initial homework grading to be reviewed by the instructor
Technology Requirements for Students	A smart phone (iPhone or Android) with a camera and “TurboScan PDF” app installed for homework scanning for submissions via LMS.

COURSE MATERIALS	
Required Books	-
Recommended Books	<i>Mathematical Methods for Physicists</i> , by Arfken, Weber and Harris <i>An introduction to Mathematical methods of physics</i> by Jones, Lorella M.
Other Materials	-
Accessing Course Materials	-

EDUCATIONAL OBJECTIVE
<i>To equip students with the advanced mathematical methods of the physical sciences and engineering – infinite series, vector and complex analysis, linear vector spaces, and differential-equation techniques – enabling them to formulate engineering and science problems in precise mathematical form and to solve them with rigor beyond the scope of the core mathematics curriculum.</i>

COURSE DESCRIPTION
<i>Topics include the principal mathematical methods underlying the physical sciences and engineering, beginning with infinite series, convergence, and Taylor approximations of physical models, and progressing through vector differential and integral calculus to the integral theorems of Gauss and Stokes. Students then study functions of a complex variable, analyticity, and the calculus of residues, and employ contour integration to evaluate definite integrals arising in applications. The treatment of linear vector spaces develops the matrix eigenvalue problem, diagonalization, functions of matrices, and the Cayley-Hamilton theorem, which are applied to coupled oscillating systems, normal modes, and Green's-function solutions of driven systems. Key practices encompass Fourier series, the Dirac delta function, special functions such as the Gamma function, and series and integral-transform solutions of ordinary differential equations. This comprehensive approach develops analytical fluency, mathematical maturity, and the ability to translate concrete physical problems into solvable mathematical structures.</i>

WEEKLY SUBJECTS (TENTATIVE)		
Week	Date	Subject
1	21-25.09.2026	Vector algebra; gradient, divergence, and curl
2	28.09-02.10.2026	Line and surface integrals; the theorems of Gauss and Stokes
3	05-09.10.2026	Infinite series: convergence tests; power series and interval of convergence
4	12-16.10.2026	Taylor expansions and their applications to physical models
5	19-23.10.2026	Complex numbers and functions; the complex logarithm and powers
6	26-30.10.2026	<i>No lecture due to Republic Day and campus closure.</i>
7	02-06.11.2026 and 07.11.2026 Saturday at 12:40 (Zoom)	Analytic functions; the Cauchy-Riemann conditions; the calculus of residues
8	09-13.11.2026	Contour integration: definite integrals, principal values, and trigonometric integrals
9	16-20.11.2026	The Gamma function; the Dirac delta function – Midterm on 20.11.2026 during lecture
10	23-27.11.2026 (Withdrawal Week)	Linear vector spaces; the matrix eigenvalue problem
11	30.11-04.12.2026 and 05.12.2026 Saturday at 14:40 (Zoom)	Gram-Schmidt orthogonalization; diagonalization; functions of matrices; the Cayley-Hamilton theorem
12	07-11.12.2026	Coupled systems: normal modes; Green's-function solutions of driven systems
13	14-18.12.2026	Power-series solutions of ordinary differential equations
14	21-25.12.2026	Function spaces and orthogonal expansions: Fourier series; Fourier and Laplace transforms

ASSESSMENT METHODS, WEIGHTS AND RULES			
Type	Weight	Implementation Rule	Makeup Rule
Final Exam	40	<i>Closed books and notes. No calculation or communication devices will be allowed during the exam.</i>	<i>Will be granted with a valid excuse.</i>
Midterm Exam	30	<i>Closed books and notes. No calculation or communication devices will be allowed during the exam.</i>	<i>Will be granted with a valid excuse.</i>
Homework	30	<i>A problem sets will be assigned after each major part. Overall weight of each problem will be proportional to the points indicated by the problem, i.e. each problem set may have different weight.</i>	<i>No makeup.</i>
Attendance/ Interaction (Participation)	0	<i>A minimum of 50% or above attendance is required to pass the course. Attendance checks will start after the add/drop period.</i>	<i>No makeup.</i>
Total	100%		

METHODS FOR TRACKING ATTENDANCE & IMPACT OF ATTENDANCE ON SUCCESS			
Method	Application Details	Required Tools and Make-up Method	Impact of Attendance of Course Grade
Signature Sheet	An attendance sheet will be circulated during class. Signing the sheet will be accepted as proof of attendance.	No technological tools are required. A pen is sufficient to sign the sheet.	At least half of the lectures must be attended when an attendance sheet is circulated. Students who fail to meet this requirement will automatically fail the course due to absence. An attendance sheet will be circulated for all lectures except for special circumstances.

ASSESSMENT(S) BEFORE THE WITHDRAWAL DEADLINE		
Type of Assessment(s)	Weight(s)	Date by When the Score of the Assessment is to be Announced
Midterm exam	30%	November 25, 2026

DETERMINING LETTER GRADE
The distribution of cumulative grades will be fitted to a gaussian curve. Grades of students who did not take the midterm or final exam or did take but received less than 5% will be excluded from this distribution before the fit. The instructor will choose the lower limit of letter grade C such that it is in between mean ± 1.5 sigma of the fit. The rest of the letter grades will be adjusted with equal spacing on the grade axis of the distribution. The final distribution will be shared on LMS without the student names after the letter grades are submitted.

EXPECTED STUDENT SEMESTER WORKLOAD	
Activities Carried Out in The Presence of An Instructor	Total Hours in Semester
Lecture (Including Discussion Sessions)	56
Recitation/Practice	0
Laboratory/Studio/Kitchen/Flight	0
Field Work	0
Activities Carried Out by The Learner Him/Herself	Total Hours in Semester
Working on Assignments	18
Pre-class Learning of Course Material	14
Review of Course Material for Exam Preparation	16
Other (after-class Learning of Course Material)	46
Overall Total Hours in Semester	150

RULES FOR THE USE OF AI IN THE COURSE	
Use of AI in the Course (Limited Use)	Provided that you comply with the Principles for the Responsible Use of AI listed below, you may use AI tools (ChatGPT, Copilot, Gemini, Claude, Grok, DeepSeek, etc.) in this course only for supportive purposes and within the restrictions stated here. You must clearly acknowledge AI contributions, act ethically and transparently, and comply with university policies. You cannot use AI to solve homework questions. You may use AI to generate similar questions to the homework questions and their solutions.
Principles for the Responsible Use of AI	Use AI as a tool to support your learning, not as the ultimate source of knowledge: AI can assist your learning and work, but the final judgment, creativity, and ownership of the output must always remain with you. Be transparent about AI's contribution: If you use AI to generate, summarize, or improve any part of your work, you must clearly acknowledge it. Transparency protects your academic honesty and strengthens professional integrity. Recognize the limitations and biases of AI: AI may produce inaccurate, incomplete, or biased outputs. Always review the results critically and verify them before using. Consult diverse sources, not just AI: Do not rely solely on AI for information. Support, compare, and cross-check AI outputs with other valid and reliable sources. Protect your data and privacy: Never share personal, sensitive, or confidential information with AI tools. Safeguarding such information is your responsibility.

OTHER RULES AND INFORMATION	
Health	The mode of instruction is subject to change in case of pandemic or other similar unavoidable reasons in line with the directives of the authorized public bodies. In such a case, the necessary revisions will be binding for all members of the OzU community, and will be announced by the relevant units, in particular Student Services.
Safety	Announced safety procedures must be strictly followed during any laboratory and similar work. Please refer to the safety guidelines posted in the laboratories. In case of emergencies, call 9911 (216 564 9911).
Accessibility	Reasonable accommodations will be provided for students with verifiable disabilities. Please inform your instructor if you need any assistance.
Academic and Technical Support	Students can get help from the Writing Center, Solution Center and IT in case they need academic and/or technical support with their classes.
Objections to Examination Results	Students who object to their exam results and cannot resolve their objections with the instructor of the course, may file a written petition for a review of their exam papers at their Dean's/Director's Office within five days from the announcement their exam results.
Academic Integrity and Honor Code	Students are assumed to abide by scholastic honesty and uphold the Honor Code. Scholastic dishonesty includes, but is not limited to cheating, attempting to cheat, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting the work of another person, or tampering with the academic work of other students. Any form of scholastic dishonesty is a serious academic violation and will result in a disciplinary action and also may result in student getting a 0 grade in the related assessment.
Unauthorized Recording/Sharing of Lectures	Recording/sharing of synchronous/asynchronous lectures or copying of lecture recordings are prohibited without the permission of the instructor. Actions such as unauthorized recording, giving the recording to someone else, taking it, using it or sharing it in physical or virtual media are prohibited and/or are offenses according to the relevant legislation. In such cases, legal procedures will be initiated in addition to starting disciplinary actions against those involved.
Flexibility	Circumstances may arise during the course that prevents the instructor from fulfilling each and every component of this syllabus; therefore, the syllabus is subject to change. Students will be notified prior to any changes.
Principles for Classroom Behavior	Creating a safe space where ideas can be freely discussed in our classroom is everyone's responsibility. In accordance with the relevant disciplinary regulations, any verbal or written expression that disrupts the order of the class, involves the unauthorized distribution of materials, or is discriminatory or violent, and that violates the dignity of individuals in the classroom (students or educators) is strictly prohibited. During class, all participants are expected to respect each other's rights.